Beyond Retrieval: Evaluating Hypothesis-Driven Reasoning in Tool-Using Agents

Anonymous authors
Affiliations and contact details to be added

Pre-results manuscript. Revised experimental design using public security datasets, 3 November 2026. No agent experiments have been run. Sample counts are recruitment targets subject to evidence audit, not confirmed dataset inventories. Appendix C specifies datasets, splits, tools, experiments, and release artifacts.

Abstract

Tool-using language agents can retrieve relevant records and execute analytical code, yet successful task completion does not reveal whether they actively test explanations. We propose an evaluation framework separating evidence acquisition, interpretation, and revision. The main study adapts public AIT enterprise logs into adjudicated investigations with entity renaming, irrelevant observations, missing decisive evidence, and delayed evidence release. OTRF Security Datasets and DARPA OpTC provide external evaluations. A separate optional regeneration study tests mechanism-changing counterfactuals; recorded-log manipulations alone are not treated as causal interventions. A hypothesis-driven agent maintains an evidence graph, chooses actions through competing predictions, and validates generated scripts. Equal-tool comparisons, common-evidence replay, and ablations test whether gains exceed those attributable to facts or computation. Primary outcomes are evidence-supported resolution and interpretation on common evidence; revision, uncertainty, script reliability, and resource use are secondary. This pre-results protocol evaluates observable behavior and makes no empirical claim of improved reasoning.

1 Introduction

A suspicious outbound transfer may reflect data theft, a scheduled backup, or incorrect attribution behind a shared network address. An agent can retrieve many relevant records and still choose the wrong explanation. The central difficulty is deciding which observation would distinguish plausible alternatives, obtaining that observation, and changing the conclusion when it contradicts an earlier belief. These behaviors are central to both security investigation and scientific inquiry.

Existing agents interleave language generation and actions [1], learn from feedback [2, 6], and integrate structured memory or learned tool policies [3–5]. These capabilities motivate a more specific evaluation question: when an investigation succeeds, what contribution comes from selecting evidence, interpreting it, and revising hypotheses? A final verdict alone cannot distinguish these mechanisms.

We propose three contributions: a matched-case evaluation protocol that manipulates decisive and irrelevant evidence; a reproducible agent architecture linking hypotheses to tool actions and validated analytical artifacts; and controlled comparisons that separate information access, procedural guidance, and evidence interpretation. These are proposed methodological contributions. Their empirical value remains to be established by the experiments specified here.

2 Related work and scope

2.1 Reasoning with actions and experience

ReAct interleaves reasoning and environment actions [1]. Reflexion uses feedback and episodic reflection to improve subsequent attempts without updating model weights [2]. ExpeL extracts transferable natural-language insights from task experience [6]. Our agent adopts an explicit external state, but stores observations separately from interpretations and records the applicability conditions of learned lessons. The proposed evaluation asks whether these additions improve evidence-sensitive behavior on held-out source environments, rather than merely improving repeated attempts at familiar tasks.

Agentic Reasoning combines web search, code execution, and a structured Mind-Map representation [3]. ReSearch trains models to integrate search within multistep reasoning using reinforcement learning [4]. ARTIST learns tool-use behavior with outcome-based reinforcement learning [5]. Our core method is an inference-time intervention with frozen model weights. It is inspired by these mechanisms but is not a reproduction of their training procedures. In particular, qualitative predictions about informative actions are not treated as calibrated Bayesian information gain.

2.2 Scientific discovery and security investigation

DiscoveryWorld evaluates complete discovery cycles in simulated environments, including hypothesis formation and experimentation [7]. DiscoveryBench examines hypothesis discovery from supplied datasets [8], while ScienceAgentBench evaluates executable programs for scientific workflow tasks [9]. The AI Scientist demonstrates a pipeline spanning idea generation, experiments, and manuscript production [10]. These works make the broad claim of autonomous science insufficient as a novelty statement. Our narrower proposed contribution concerns controlled attribution of performance and response to evidence interventions.

ExCyTIn-Bench evaluates multihop investigation across security logs [11], and CFA-Bench uses packet captures and logs for forensic investigations [12]. They provide relevant task designs and potential external validation settings. We do not assume that their existing tasks isolate hypothesis revision, knowledge-matched playbooks, or validated script generation. Any reuse must preserve their evaluation contracts and clearly distinguish original tasks from new variants.

Table 1. Positioning relative to selected prior work. Descriptions summarize scope, not an exhaustive feature audit.

Work	Primary emphasis	Role in this study
ReAct and Reflexion [1, 2]	Action loops and feedback	Agent and reflection baselines
Agentic Reasoning [3]	Tools and structured memory	Graph-memory motivation
ReSearch and ARTIST [4, 5]	Learning search and tool policies	Future training direction
ExpeL [6]	Experience-derived guidance	Conditional lesson memory
DiscoveryWorld [7]	Scientific discovery cycles	Intervention-based task design
ExCyTIn and CFA-Bench [11, 12]	Cybersecurity investigations	Domain evaluation context

2.3 Claims the study can support

We use “reasoning” operationally to mean evidence-sensitive selection, composition, and revision of explanations. A behavioral advantage would not establish human-like cognition, faithful internal reasoning traces, or general scientific competence. Cybersecurity is the primary domain. Broader claims require replication in at least one non-security domain with comparable interventions.

3 Investigation as sequential hypothesis testing

3.1 Task and observable state

Each task specifies an initial question, a partially observed environment, an available tool registry, and a resource budget. The evaluator retains the scenario mechanism, a complete event history, and an adjudicated evidence rubric. The agent receives only the initial alert and authorized observations. At each step it chooses a query, analysis, script execution, or permitted sandbox probe, observes the result, and updates its external investigation record. It terminates with a verdict, affected entities, a supported explanation, calibrated confidence, and unresolved questions.

The physical scenario label and the epistemically justified answer are distinct. A compromised host can remain unresolvable from the available evidence. The answer space therefore includes malicious, benign, and insufficient evidence. The last label is assigned by an explicit observability rubric rather than inferred from the absence of detections. Ground truth can admit multiple supported explanations when the evidence cannot distinguish them.

3.2 Hypothesis-driven agent

The proposed agent, HDA, maintains an evidence graph with four node types: immutable observations, hypotheses, analytical artifacts, and unresolved questions. Edges encode support, contradiction, derivation, and dependency. Observations retain source identifiers, timestamps, query parameters, and hashes. Hypotheses retain scope and status. Derived findings point to both their input records and the exact script version. A graph edge is an agent assertion until independently verified; its presence is not evidence that the relation is correct.

Before an information-gathering action, HDA records a compact action contract: the question being tested, at least two plausible explanations when ambiguity exists, predicted observations under each, the selected tool and parameters, and how the result could change the conclusion. Predictions must be distinguishable; “the tool will provide useful information” is invalid. The agent may retain an open-set alternative rather than force the true explanation into an incomplete list. A validator checks structure, not semantic truth, and all repair attempts consume budget.

After execution, the agent checks observation quality and updates only supported relations. Failed tools, empty results, truncated results, and valid negative observations have separate statuses. Before dismissal or escalation, the agent attempts one available discriminating check against its leading explanation, or records why no such check is feasible. This requirement is evaluated for usefulness, not merely for the appearance of a skeptical sentence.

3.3 Scripts and lesson memory

Custom scripts run in an isolated workspace against the same evidence snapshots available to other conditions. A script contract names its inputs, join keys, units, expected output schema, and assumptions. Validation uses a small known-answer fixture plus an independent check of sampled records or a reference operation. A script that runs but fails validation cannot contribute a verified finding. Fixture creation, debugging, and checking all count toward the agent budget; scripts never receive evaluator labels or hidden evidence.

Experience memory is constructed only from training cases. Each lesson stores an applicability condition, a recommended test, known failure conditions, and training-case provenance. The memory is frozen before test evaluation. No lessons are shared across test cases. We compare the full method against removal of each component and a length-matched flat-memory control.

4 Benchmark construction

4.1 Public datasets and evaluation units

The core corpus is derived from AIT Log Data Set V2.1 [16], with external cohorts from OTRF Security Datasets [18] and DARPA OpTC [19]. The target is 40 AIT training cases, 40 development cases, and 80 test cases, separated by complete source environments. Cases are analyst-authored investigation questions grounded in original records; they are not pre-existing benchmark questions. Appendix C defines the exact source allocation and eligibility rules. Counts may decrease after audit. No claim is made that the datasets contain 24 independent incident-mechanism families.

An investigation case consists of a focal question, a fixed time interval, accessible telemetry, and an independently authored evidence rubric. Multiple questions drawn from one campaign remain one dependence cluster. Split environments before writing questions or learning lessons; do not split events randomly. The core AIT evaluation measures transfer across environments with potentially shared mechanisms. OTRF and OpTC assess external dataset transfer, not guaranteed novelty relative to model pretraining. Verified benign cases require affirmative contextual support; lack of an attack label is not sufficient.

Table 2. Recorded-data variants and the separate optional regenerated counterpart. Eligibility is determined before agent runs; not every base case supports every variant.

Version	Controlled change	Expected response
V0 Original	Original public records; frozen case scope	Evidence-supported resolution
V1 Renaming	Consistent opaque entity identifiers across all sources	Same answer after mapping
V2 Distractors	Additional unrelated activity from same recording	Stable supported conclusion
V3 Regenerated optional	Mechanism changed in a new paired testbed run	Correct alternative explanation
V4 Missing evidence	All access paths to a mandatory finding removed	Explicitly qualified answer
V5 Delayed evidence	Shared incomplete bundle, then decisive release	Appropriate revision or resolution

4.2 Valid interventions and evidence audit

The core study uses V0, V1, V2, V4, and V5 on recorded data. V1 applies a consistent entity mapping; V2 expands the accessible view with unrelated same-environment activity; V4 removes every permitted route to a required conclusion; V5 releases previously withheld evidence at a common checkpoint. None changes what physically occurred. V3 is reserved for a separately generated, physically coherent paired scenario using the open Kyoushi testbed [17]. Unrelated recordings must not be called counterfactual pairs. Two reviewers verify evidence sufficiency and variant validity before test execution.

In V5, all policies first receive the same deliberately incomplete bundle, provide a provisional structured answer, and then receive the same decisive bundle before an equal continuation budget. The benchmark targets evidence revision, without forcing an incorrect initial conclusion. Qualified initial answers are valid. All eligible cases contribute to final ESR; initially wrong cases form a secondary correction analysis. A separate control releases non-decisive late evidence to detect indiscriminate answer changes. Natural adaptive discovery of contradictions is exploratory and reported separately.

Every run receives a fresh read-only evidence snapshot. Agent-facing mounts exclude labels, attack scripts, scenario titles, public walkthroughs, evaluator code, and analyst notes. Normalize case identifiers and filenames, but retain operational semantics such as event types and meaningful command arguments. Derived alerts that explicitly name the attack are excluded from the primary raw-evidence track. Public-corpus memorization cannot be ruled out by entity renaming; report a no-evidence diagnostic and distinguish public-record performance from the optional newly generated cases.

5 Experimental conditions and controls

Table 3. Condition definitions. A, H, AS, and HS comprise the core grid; R and P are secondary controls on a preregistered subset.

ID	Condition	Adaptive queries	Custom scripts	Guidance
R	Fixed retrieval plus synthesis	No	No	Common domain facts
A	Standard tool-using agent	Yes	No	Common domain facts
P	Agent with fixed playbook	Yes	No	Same facts in ordered steps
H	Hypothesis-driven agent	Yes	No	Graph and action contracts

ID	Condition	Adaptive queries	Custom scripts	Guidance
AS	Standard agent with scripts	Yes	Yes	Common domain facts
HS	Full hypothesis-driven agent	Yes	Validated	Graph, contracts, lessons

R uses a deterministic, development-tuned retrieval plan followed by the same model used in other conditions. Its synthesis stage may reason over evidence; “retrieval baseline” does not mean a reasoning-free model. A uses a conventional action–observation loop [1]. P receives the same domain propositions as A, rearranged into procedural guidance. H receives hypotheses and graph structure but no custom-code access. AS and HS provide the key comparison with equal tool expressivity. The lesson component is separately removed in the ablations.

5.1 Knowledge and computation matching

A common reference contains every domain fact used by playbooks and learned lessons, excluding incident-specific answers. Reviewers map each guidance statement to this reference. Procedures add ordering or decision rules, not extra privileged facts. Context-length controls use an equally sized flat record of the same content. All conditions use the same query schemas, pagination policy, time-zone normalization, package environment, and access to data. Script execution is the only intentional capability difference in the script comparisons.

Use two frozen model backbones from different families, with exact versions and inference settings selected before test execution. The default run cap is 40 tool invocations, 30,000 generated tokens where measurable, and 20 minutes wall-clock time; script CPU time is limited to 120 seconds per call. All repairs and validation attempts consume budget. Record input and output tokens, hidden reasoning tokens when exposed, CPU time, wall time, and actual currency cost. These caps are pilot defaults, not power or cost guarantees. Common evidence uses a 12,000-token input ceiling and a 4,000-token output ceiling shared by both synthesis policies; cases whose decisive content cannot fit are excluded from replay before running models.

5.2 Common-evidence and oracle diagnostics

To isolate interpretation, every synthesis policy receives identical evidence bundles with identical ordering and provenance. The principal replay uses bundles selected independently by a fixed development-frozen collector. A supplementary cross-over replay uses bundles from A and HS: both interpreters process each bundle without the collecting agent’s hypotheses, predictions, or conclusions. This cross-over estimates interpretation differences conditional on those bundles; it is not an unbiased decomposition of total agent performance because evidence selection is itself affected by the agent.

An oracle-evidence condition supplies a complete decisive bundle for resolvable cases. It diagnoses residual interpretation failures and is not assumed to be an upper bound. Oracle inputs are excluded from missing-evidence tasks. Any gain from extra synthesis passes is compared against a matched-budget reflection baseline. All collection and interpretation costs are retained, including unsuccessful retries.

6 Outcomes and adjudication

6.1 Primary outcomes

Evidence-supported resolution (ESR) is a binary case-level outcome. A run passes only if it gives an admissible verdict, recovers all case-specific mandatory findings, cites evidence that entails those findings, and contains no material unsupported claim. Rubrics identify mandatory findings before model runs and do not require a particular tool sequence. A finding is material if it changes the alleged mechanism, scope, affected entity, or recommended disposition. Empty answers fail; justified insufficient-evidence answers can pass V4.

Counterfactual pair success (CPS) is defined only for the optional V0/V3 regenerated cohort: both members must achieve ESR under paired run indices. It is not computed for unrelated public recordings. The core primary

comparisons use ESR, with V0 analyzed separately from each intervention. Renaming and distractor robustness use pair success after canonical entity mapping; raw agreement is secondary because consistently wrong answers are not robust. Report per-variant denominators and do not average ineligible variants as failures.

Table 4. Outcome definitions and scoring sources.

Metric	Operational definition	Scoring source
ESR ↑	Admissible answer, required findings, valid support, no material invention	Frozen evidence rubric
CPS ↑	Both original and counterfactual pass ESR	Paired run outcomes
Revision ↑	Post-release ESR; report provisional state and direction of update	Shared evidence checkpoint
Finding precision / recall ↑	Correct unique claims / asserted claims; recovered / required facts	Atomic claim audit
Unsupported claims ↓	Unsupported material claims per run; also fraction of all claims	Blinded adjudication
Calibration ↓	Three-class Brier score on evaluable answer labels	Recorded probabilities
Valid script yield ↑	Unique verified findings contributed by validated scripts per run	Fixtures and record checks
Cost ↓	Calls, tokens, latency, CPU seconds, currency cost	Execution traces

6.2 Action quality and uncertainty

On a balanced, preregistered subset of decision checkpoints, reviewers assess whether an action can distinguish currently plausible hypotheses, whether its predicted outcomes are meaningful, and whether an equally accessible lower-cost action would answer the same question. We report this as action informativeness, not measured information gain. No credit is assigned because a tool name looks sophisticated. Query utility is additionally measured through newly recovered required findings per call, with repeated findings deduplicated.

Confidence is elicited as probabilities over the three answer categories at fixed checkpoints and at termination. Abstention quality is reported together with coverage and selective risk; high abstention alone is not success. Missing-evidence tasks are analyzed separately from tasks with ordinary low confidence. Revision is evaluated on every delayed-evidence task, with initially wrong cases shown as a secondary conditional analysis to avoid selection bias.

Two blinded reviewers score evidence support and action quality using shuffled outputs with condition labels removed. A frozen LLM judge may assist claim segmentation but does not supply the primary truth labels. Report reviewer agreement, adjudication rates, and the number of corrected labels. Ties, ambiguous entailment, and evaluator failures follow a published adjudication policy.

7 Hypotheses and statistical analysis

Table 5. Confirmatory hypotheses and contrasts. Higher ESR and CPS indicate better performance.

Hypothesis	Contrast	Primary endpoint
H1 Structure aids investigation	H versus A	V0 ESR
H2 Structure aids interpretation	H-style versus A-style synthesis on fixed bundles	ESR
H3 Full method exceeds equal tool access	HS versus AS	ESR

Hypothesis	Contrast	Primary endpoint
H4 Validation adds value	HS versus HS without script validation	V0 ESR on script-useful subset

H1–H4 are the confirmatory family for the recorded-data study. H1 and H3 use V0 ESR; H2 uses fixed-bundle ESR; H4 uses the preregistered script-useful V0 subset. Variant effects and external transfer are secondary. CPS is secondary in the optional regeneration study. Freeze scoring, effect directions, and analysis code before test runs. If clustering prevents a sufficiently precise confirmatory analysis, label the study exploratory before unsealing outputs rather than presenting the nominal run count as statistical power.

7.1 Sample size and repeated runs

The ceiling for the core grid is 80 AIT test cases × 5 recorded-data versions × 4 conditions (A, H, AS, HS) × 2 backbones × 3 repeats = 9,600 runs, assuming every case supports every version. The actual count is 24 times the sum of eligible case-version combinations. Fixed-bundle replay adds 960 synthesis runs; a 40-case validation ablation adds 240 new runs by reusing the matched HS runs; two external 24-case cohorts add 1,152 runs. Thus the planned ceiling is 11,952 runs, excluding pilots and optional extensions. Appendix C gives allocation formulas. Repeats and variant transformations are not independent incidents. Use pilot simulations for detectable effects; do not assert power to detect a five-percentage-point gain from four held-out environments.

7.2 Estimation and multiplicity

Average repeats within case and condition, compute paired differences, and weight cases equally for the main descriptive estimate. Report environment-macro averages and each environment separately. Keep all variants and repeats together in resampling; cluster related questions by their originating campaign or simulation. Report case-cluster intervals conditional on the sampled environments and leave-one-environment-out sensitivity. Only four core test environments are available, so population-level uncertainty remains poorly estimated. Do not use a narrow case bootstrap to claim generalization across organizations or mechanisms.

Apply Holm correction across H1–H4 separately for each backbone. If a confirmatory paired randomization test is used, swap condition labels at the independent campaign/environment cluster level, not at the repeat or overlapping-window level. With only four independent test environments there are only 16 binary swap assignments, making conventional significance thresholds generally unattainable, especially after multiplicity correction. Report this limitation directly. Broader confirmatory claims require additional independent source environments or newly generated independent campaigns registered in advance. Report effect sizes and intervals regardless of significance, and retain the distinction between finite-corpus results and population inference.

7.3 Failure handling and reproducibility

Agent-induced invalid calls, exhausted budgets, and generated-code errors remain failures. Verified infrastructure outages are rerun under the same configuration and documented; no output-dependent exclusions are allowed. Unresolved outages are reported as missing with worst-case sensitivity bounds. Case exclusions for flawed ground truth require blinded review and disclosure for all affected conditions. We will publish configuration hashes, prompts, environment snapshots, query traces, script versions, validator outcomes, confidence reports, and scoring scripts subject to data licensing and privacy constraints.

8 Results and planned analyses

No experiments have been executed for this manuscript. The tables below are reporting templates. "—" means not measured, not zero. Each table must be populated separately for both frozen model backbones; no hypothetical numbers or simulated trends should be substituted for empirical findings.

8.1 Main investigation outcomes

Table 6. Reporting template. Populate A, H, AS, and HS for the full V0 cohort; show R and P only on their shared secondary subset, with denominators. Report ESR and uncertainty by backbone; CPS belongs to the separate regenerated cohort.

Condition	ESR ↑	CPS ↑	Unsupported claims ↓	Calls ↓	Cost / run ↓
R Fixed retrieval	—	—	—	—	—
A Standard agent	—	—	—	—	—
P Fixed playbook	—	—	—	—	—
H Hypothesis agent	—	—	—	—	—
AS Agent + scripts	—	—	—	—	—
HS Full method	—	—	—	—	—

The main comparison is HS versus AS, which holds custom-script access constant. H versus A tests structure without scripting. P versus A tests procedural guidance with shared domain knowledge. Report whether improvements persist after accounting for actual compute use and whether they are consistent across incident families. Do not attribute HS gains to hypothesis testing if they disappear in the relevant ablation or arise solely from a higher execution budget.

Figure 1. Performance and resource use. Two panels: ESR by condition with paired uncertainty intervals; ESR versus median total cost at preregistered budget settings. Show measured points only. Use a consistent condition legend and separate markers for each backbone.

8.2 Reading null and mixed outcomes

If H improves acquisition but not common-evidence interpretation, the supported claim concerns search policy rather than general inference. If AS and HS perform similarly, expressive tool access may explain most gains. If P succeeds only on familiar mechanisms, procedural compliance is an inadequate account of transfer. If effects are small or confidence intervals wide, report the uncertainty without converting secondary metrics into replacement primary endpoints.

8.3 Robustness and hypothesis revision

Table 7. Intervention results template. Core results cover renaming, distractors, missing evidence and delayed evidence. Counterfactual CPS is reported only for the optional regenerated cohort; P appears only on its secondary subset.

Condition	Rename pair ↑	Distractor pair ↑	Counterfact. pair ↑	Missing ESR ↑	Revision ESR ↑
A Standard agent	—	—	—	—	—
P Fixed playbook	—	—	—	—	—

Condition	Rename pair ↑	Distractor pair ↑	Counterfact. pair ↑	Missing ESR ↑	Revision ESR ↑
H Hypothesis agent	—	—	—	—	—
AS Agent + scripts	—	—	—	—	—
HS Full method	—	—	—	—	—

For delayed contradictions, report the number of initially correct, wrong, and unresolved provisional answers in every condition. Show corrections and regressions separately. A model that changes its answer indiscriminately should not receive credit for revision. Include misleading but non-decisive late evidence as an exploratory control to distinguish appropriate updating from recency bias.

PLOT SPACE — NO EXPERIMENTAL DATA

Figure 2. Sensitivity to decisive and irrelevant evidence. Plot V1/V2 pair success and V4 ESR with intervals and denominators; show optional V3 CPS separately. Report benign and malicious cases separately.

PLOT SPACE — NO EXPERIMENTAL DATA

Figure 3. Revision after evidence release. Use provisional and final probability distributions at the fixed checkpoint. Plot successful corrections and harmful reversals separately; include all delayed-evidence cases and disclose conditional-subset denominators.

8.4 Attribution and component ablations

Table 8. Common-evidence interpretation template. Each row is a paired comparison on identical observations; collector thoughts and conclusions are removed.

Evidence source	Standard ESR	Structured ESR	Paired Δ [95% CI]
Independent fixed collector	—	—	—
A-collected bundle	—	—	—
HS-collected bundle	—	—	—
Oracle decisive bundle	—	—	—

Table 9. Ablation template. Match content, available tools, and budget; record actual resource differences.

HS variant	ESR ↑	CPS ↑	Revision ↑	Cost ↓
Full method	—	—	—	—

HS variant	ESR ↑	CPS ↑	Revision ↑	Cost ↓
Flat memory with same content	—	—	—	—
No pre-action predictions	—	—	—	—
No contradiction check	—	—	—	—
No script validation	—	—	—	—
No experience lessons	—	—	—	—
Fixed procedural lessons	—	—	—	—
Matched-budget reflection baseline	—	—	—	—

The flat-memory comparison preserves evidence and hypothesis text while removing graph organization. The no-predictions condition preserves the same action loop and memory. The no-validation condition can run the same generated scripts but does not enforce the validation gate. Its evaluation still checks findings independently. A repeated-review baseline receives the same token budget as HS, helping distinguish structured investigation from additional opportunities to reconsider.

PLOT SPACE — NO EXPERIMENTAL DATA

Figure 4. Component effects and interpretation. Plot paired ESR differences between HS and each ablation, with a vertical zero line and clustered confidence intervals. Add a separate panel for common-evidence comparisons. Do not label the cross-over panel as a causal mediation estimate.

8.5 Custom scripts and failure analysis

Table 10. Script evaluation template. Report conditional script statistics alongside unconditional case-level ESR.

Condition	Runs using scripts	Execution success	Validation pass	Verified findings/run	False findings/run
AS Standard scripts	—	—	N/A	—	—
HS without validation	—	—	N/A	—	—
HS Full method	—	—	—	—	—

Measure script contributions on tasks marked script-useful by an independent rubric before evaluation. Do not select tasks retrospectively because HS chose to write code. Useful scripts may normalize timestamps, correlate identities across sources, reconstruct event sequences, or compute aggregate statistics. All conditions retain access to the same raw inputs. Countercheck derived findings against immutable records, and distinguish a correct implementation of a wrong analytical assumption from a programming error.

Figure 5. Script reliability and evidence yield. Plot the counts of attempted, executable, validated, and evidence-producing scripts. Because these events are nested only in the validated condition, show other conditions separately. Add verified findings per run versus script CPU time, including zero-yield runs.

8.6 Failure taxonomy

Assign each failed run a primary bottleneck using the first outcome-changing error supported by its observable trace: missed evidence; incorrect entity or time alignment; unsupported inference; untested alternative; failure to revise; invalid script semantics; premature stopping; or environment interaction failure. Permit secondary labels, but count the primary categories once per failed run. Reviewers should not infer hidden cognition from verbal explanations. Present representative successes and failures selected by a preregistered sampling rule, with paired cases rather than isolated favorable anecdotes.

A useful negative result is that structured contracts increase overhead without improving ESR. Another is that script validation improves precision but reduces coverage under tight budgets. Either result would identify a practical boundary of the method. Report these tradeoffs directly rather than presenting a single aggregate score that obscures them.

8.7 Optional investigation graph experiments

An investigation graph is a representation of observed events and asserted evidential relations, not a ground-truth causal graph. Graph of Thoughts organizes generated intermediate representations [13]; our proposed graph additionally distinguishes immutable records, inferred relationships, and competing incident explanations. We test whether this distinction improves research performance beyond presenting the same facts as text.

8.7.1 Representation and graph construction

Use a two-stage design. In the representation experiment, an independent collector supplies a common evidence bundle. Every condition receives the same entities, event content, candidate hypotheses, and relation propositions. Render these as a chronological ledger, a relation list, or a typed graph. Match the context budget without adding irrelevant padding and report residual token differences. The ledger retains relation statements in prose so graph structure does not introduce new facts. The graph is serialized as typed node and edge records; no visual graph-reading capability is required.

In the construction experiment, each agent builds its representation from raw tool outputs. Compare an entity-and-time graph, a graph with support and contradiction links, and the full provenance-aware graph with explicit unresolved questions. All construction, updating, and retrieval calls count toward the same budget. Evaluate relation precision, relation recall against the adjudicated case rubric, and preservation of contradictory evidence. Equivalent supported relations receive credit even when the graph differs from a reference topology.

Table 11. Investigation graph results template. Run both common-evidence and online-construction settings; report them separately.

Representation	ESR ↑	CPS ↑	Relation precision ↑	Token cost ↓
Chronological evidence ledger	—	—	—	—
Flat list of identical relations	—	—	—	—
Entity and temporal graph	—	—	—	—
Support and contradiction graph	—	—	—	—

Representation	ESR ↑	CPS ↑	Relation precision ↑	Token cost ↓
Full graph with provenance and gaps	—	—	—	—

8.7.2 Incorrect and missing relations

In a separate robustness experiment, corrupt the external working memory while keeping raw observations unchanged. Remove or swap 0%, 10%, and 20% of eligible relation edges using seeded procedures that preserve node counts and edge types. Mark all derived relations as unverified in every condition; do not disguise corrupted content as trusted ground truth. Ask whether the agent detects inconsistent relations by checking source evidence. Apply an equivalent relation corruption to the flat-list control. Report repaired relations, propagated false claims, and ESR as a function of corruption rate. These rates are experimental design choices, not observed error frequencies.

PLOT SPACE — NO EXPERIMENTAL DATA

Figure 6. Graph utility and resistance to incorrect relations. Left: paired ESR differences under common evidence. Right: ESR and false-claim rate against edge-corruption percentage, with matching flat-list controls. Show graph construction cost separately from downstream query cost.

8.8 Optional techniques for improving research performance

We compare five additional interventions that target planning, review, and experience reuse. Tree of Thoughts motivates branching over candidate solutions [14], while multiagent debate motivates independent proposals and critique [15]. Their published results do not establish cybersecurity benefits. Our implementations are domain adaptations and must be labeled as such, rather than claimed as exact reproductions.

Table 12. Technique comparison and results template. All methods use the same backbone and permitted evidence; dashes denote unmeasured outcomes.

Technique	Controlled intervention	ESR ↑	CPS ↑	Cost ↓
Hypothesis tree	At most three branches; evidence-based pruning and backtracking	—	—	—
Independent critique	Fresh-context reviewer challenges evidence support before finalization	—	—	—
Retrieval-guided planning	Retrieve frozen training lessons by current evidence gap	—	—	—
Adaptive replanning	Revise the plan after contradiction, failed action, or no new finding	—	—	—
Independent candidates	Three proposals on the same evidence; evidence-based adjudication	—	—	—
Matched self-review	Single agent revisits its answer using equivalent aggregate budget	—	—	—

Independent critique receives the evidence bundle and proposed claims without the authoring agent's rationale. The critic must cite contradictory or missing evidence; disagreement alone earns no credit. Independent-candidate

aggregation selects evidence-supported conclusions rather than majority vote. Hypothesis-tree branch scores use validated observations rather than the model's confidence alone. Adaptive replanning is compared with a fixed plan containing the same allowed actions, while lesson retrieval is compared with a length-matched fixed lesson set and a factual reference containing the same domain propositions.

8.8.1 Budgets and interaction effects

These additional techniques are exploratory and are not included in the 11,952-run ceiling. Select at most one graph representation comparison and one critique comparison using development data before reserving a fixed test subset. A separate optional regenerated cohort of 24 pairs would add 576 runs (24 × 2 members × 2 conditions A/H × 2 backbones × 3 repeats). Do not expand the primary scope after observing outcomes or pool regenerated data with historical recordings.

These comparisons are secondary and do not expand H1–H4 after observing results. Freeze all selected combinations on development data; retain negative and null technique results. Define research performance as supported resolution, useful hypothesis tests, appropriate revision, reproducibility of derived findings, and efficiency. Report whether improvements persist at both lower and higher budget caps. The expanded study will require additional runs beyond the 11,952-run planned ceiling; its subset allocation and costs must be registered before execution.

Figure 7. Research performance under alternative techniques. Plot ESR against total cost for each technique at fixed budget levels, with uncertainty intervals. Add the graph-by-critique interaction estimate and interval; do not draw a trend or Pareto frontier before collecting observations.

9 Discussion and limitations

9.1 What would count as evidence beyond retrieval

A strong empirical pattern would combine correct counterfactual discrimination, stable conclusions under irrelevant changes, appropriate uncertainty when decisive evidence is absent, and an interpretation advantage on common evidence. This combination would support the usefulness of hypothesis-driven structure for observable investigative behavior. It would not establish that external action records faithfully expose an internal reasoning process. Correct predictions may be generated after familiar-pattern recognition, and controlled variants only reduce, rather than eliminate, such explanations.

Playbooks may reduce omissions without improving transfer. Evidence graphs may function primarily as memory compression. Generated scripts may add computational capability without improving judgment. The proposed controls are intended to distinguish these narrower accounts. The study should report the smallest claim justified by the contrasts, including null results and disagreement across backbones. We do not claim that the architecture increases the intrinsic reasoning capability of a frozen model.

9.2 Scientific inquiry and external validity

Most tasks reconstruct events from observational data. This is closer to abductive investigation than unrestricted scientific discovery. Sandbox probes and regenerated counterfactuals introduce experimental structure, but causal conclusions require explicit assumptions and valid interventions. An agent querying another log source has acquired evidence; it has not necessarily intervened on the mechanism. Claims about autonomous science should therefore remain limited to hypothesis testing and evidence revision in the evaluated setting.

The public datasets are historical and differ in realism, telemetry coverage, and labeling quality. AIT includes simulated environments with shared attack mechanisms; OTRF atomic captures may lack sufficient benign context; OpTC contains documented telemetry defects; LANL, if used, has incomplete malicious-event annotation. Four held-out AIT environments limit population inference. Public examples may be represented in pretraining even after renaming. Case counts, transformation counts, and repeated runs must not be mistaken for independent

mechanisms. Additional limitations include analyst assumptions in rubrics and undetected semantic errors in generated scripts.

9.3 Responsible evaluation

The study uses isolated, resettable environments, synthetic identities, and read-only investigation interfaces by default. Any active probe is restricted to the approved range and its effects are logged. Raw incident data should be de-identified before external release. Tool outputs are treated as untrusted evidence, and a separate exploratory test can measure susceptibility to instructions embedded in logs; those results should not be mixed into the primary reasoning benchmark. The agent does not autonomously remediate production systems in this protocol.

10 Conclusion

Beyond Retrieval specifies a public-data evaluation of hypothesis-driven investigation. The core experiments separate acquisition from interpretation and test sensitivity to missing, irrelevant, and newly released evidence. The method combines provenance-aware graphs, action predictions, validated scripts, and conditional lessons. Optional regenerated pairs evaluate mechanism changes separately. Until experiments are completed, the manuscript offers an implementable protocol rather than an empirical improvement claim. The decisive question is whether gains survive equal-information and equal-budget controls, and whether their scope is supported by the independent environments actually evaluated.

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Appendix A Worked task and agent contract

The following is an illustrative benchmark specification, not an observed result. It demonstrates how to construct a testable investigation without encoding the answer in the initial alert. This outbound-transfer example is conceptual; include it in a public-data cohort only if the necessary attribution and authorization records are actually present.

A.1 Outbound transfer with ambiguous attribution

Initial question: “An outbound transfer exceeded the usual volume from a server segment. Determine whether the observed transfer is authorized activity, a compromise-related transfer, or unresolved from available evidence. Identify the originating host and support the conclusion.” Available sources include network flow, address-translation records, endpoint process events, authentication logs, and job configuration snapshots.

Competing explanations are an authorized backup, unauthorized data staging and transfer, and incorrect attribution caused by address translation. Decisive evidence must jointly establish the internal host, initiating process, and authorization context. An external destination reputation label alone is insufficient. In the counterfactual, change the initiating process and authorization mechanism, then regenerate dependent telemetry. In the missing-evidence case, remove the translation evidence and any alternative source that identifies the host, preserving genuine attribution ambiguity.

Table A1. Minimal structured output for an investigation step.

Field	Required content
Hypotheses	IDs, descriptions, unresolved alternatives, and current status
Question	Specific uncertainty that affects the final answer
Predictions	Different expected observations under competing explanations
Action	Tool, parameters, scope, and resource estimate
Observation	Status, immutable source IDs, content hash, and timestamps
Update	Supported changes, contradictions, and remaining gaps
Script artifact	Version, inputs, assumptions, validation outcome, and output IDs
Final answer	Verdict probabilities, required findings, citations, and limitations

A.2 Reference execution procedure

1. Initialize the permitted tools, frozen lesson set, and empty evidence graph.
2. Register plausible hypotheses, including an unresolved alternative when needed.
3. Identify an outcome-changing uncertainty and select an action with discriminating predictions.
4. Validate action structure, execute within the budget, and retain raw output provenance.

5. If code is generated, validate its semantics before admitting derived findings.
6. Update hypothesis status; retain contradictory observations and unresolved gaps.
7. Attempt an available falsification check before final disposition.
8. Stop at sufficient evidence or budget exhaustion and produce the structured report.

These records are concise observable artifacts for audit and scoring. They are not requests for hidden chain-of-thought, and verbosity itself receives no reward. The final grading checks evidence and decisions independently of the agent's explanation of why it acted.

Appendix B Reproducibility and completion record

Table B1. Configuration fields to freeze before confirmatory evaluation.

Field	Planned value or required entry
Dataset allocation	AIT: 40 train / 40 development / up to 80 test cases across 2 / 2 / 4 whole environments; OTRF and OpTC: up to 24 external cases each.
Primary test grid	Up to 80 × 5 eligible versions × 4 conditions × 2 backbones × 3 repeats = 9,600 core runs. See Appendix C for other allocations.
Backbone identifiers	[Exact provider/model version and release identifier]
Inference settings	[Temperature, top-p, reasoning effort, context and output caps]
Initial run caps	40 tool calls; 30,000 generated tokens; 20 minutes wall clock
Script execution	120 CPU seconds per call; [memory cap and package lock hash]
Evidence policy	[Snapshot hashes, tool schemas, pagination, truncation rules]
Guidance content	[Common reference, playbook and frozen lesson-set hashes]
Evaluation	[Rubric version, reviewer protocol, metric implementation hash]
Randomization	[Case order, paired run-index scheme, environment seeds]
Power decision	[Development-only simulation, final sample size and rationale]
Registration	[Timestamped protocol URL and deviations log]

B.1 Data and artifact records

Each run record should contain case and family IDs; variant; condition; backbone version; randomization seed or documented provider limitation; start and stop timestamps; budget usage; tool calls and statuses; immutable evidence references; script contents and hashes; validation checks; provisional and final answers; scorer decisions; and exclusion or rerun reasons. Preserve raw traces as well as canonicalized evaluation views so changes in preprocessing can be audited.

B.2 Submission completion requirements

Before submission, replace every unmeasured table cell and plot space with verified outputs; freeze model and environment metadata; complete power and uncertainty analyses; document corpus construction and exclusions; release permitted artifacts; and rewrite the abstract and conclusion around actual findings. Add funding, author contributions, and any required ethics or reproducibility statements. Select a target conference and apply its official

formatting and page limits. This draft uses an editable single-column manuscript layout and is not an official conference template.

B.3 Optional extensions outside the primary claims

A subsequent training study could use outcome-based reinforcement learning to optimize action selection, motivated by ReSearch and ARTIST. Such a study must separate additional training data, training compute, and inference-time structure. A second-domain evaluation could use controlled scientific simulations to test whether hypothesis contracts transfer beyond cybersecurity. Neither extension is included in the confirmatory results proposed here; both require their own preregistered controls.

Appendix C Experiments using public datasets

C1 Dataset selection and permitted claims

The proposed study uses existing public telemetry rather than requiring a new cyber range for its core experiments. Source documentation and license metadata were checked on 3 November 2026; the full archives have not been downloaded or audited in this design exercise. Availability of a source is not evidence that all proposed case quotas can be met. “Public” and “open source” are not interchangeable: AIT is research-usable under a noncommercial share-alike license, whereas the selected OTRF repository uses MIT and OpTC carries a public-domain release statement.

Source	Study role	Evidence and qualification
AIT LDS V2.1 [16]	Core experiments	Multi-host logs, simulated normal activity, separate attack labels. Use no-PCAP archives first. CC BY-NC-SA 4.0.
OTRF Security Datasets [18]	External transfer	Windows technique recordings. Admit only questions supported by actual source coverage. MIT at repository level.
DARPA OpTC [19]	External transfer	Structured endpoint provenance and red-team ground truth. Apply documented errata; public-domain release.
LANL cyber1 [20]	Optional scale study	Authentication, process, DNS and flow records. Red-team events are positive annotations, not exhaustive benign labels.
Splunk BOTSv3 [21]	Optional replication	CC0 repository; pre-indexed distribution requires a verified extraction path. Not a core dependency.

The main study excludes feature-only intrusion-classification tables because they generally cannot support provenance-based investigation questions. LANL is reserved for entity linking, event reconstruction and scaling; do not claim process-command explanations that its de-identified fields cannot establish. BOTSv3 is reserved because its standard preparation route uses Splunk software and public challenge answers create contamination risk. No optional source is needed to execute the core AIT design.

C2 Fixed source allocation

Partition	Source environments	Target base cases
Training	AIT fox and harrison	40 total; up to 20 each
Development	AIT russellmitchell and santos	40 total; up to 20 each
Core test	AIT shaw, wardbeck, wheeler and wilson	80 total; up to 20 each
External test	OTRF and OpTC; no lesson learning	24 each; distinct source clusters where feasible

This allocation is a proposed freeze, not a claim of mechanism-disjoint splitting. Entire archives, case variants, related hosts and campaign artifacts remain in their partition. AIT versions and derived AIT alert/netflow releases are the same source lineage, not new independent datasets. Tune parser code using schemas and development fixtures; do not tune prompts on external test answers. An independent curator may inspect test records to build rubrics while model developers remain blind to them.

C3 Case construction and evidence rubrics

For each AIT test environment, target ten attack-related investigations and ten positively supported benign investigations. Balance by scenario rather than by individual log lines. This quota is conditional on evidence: do not invent benign authorization records or force ambiguous cases into benign labels. Exclude candidates that require hidden operator intent unavailable in the agent evidence. Keep a candidate registry recording all admissions and exclusions before model runs.

Start from a documented event episode and create a neutral question that requests the involved entity, event sequence, mechanism supported by records, and remaining uncertainty. Aim for two or more relevant telemetry sources and at least two plausible initial explanations. A default 60-minute interval is a pilot setting; expand to encompass the full episode and context using a fixed, development-tested rule. The task declares the accessible scope. A question count never implies an equal count of independent attacks.

Candidate question	Candidate evidence	Admission rule
Did a web request lead to server-side execution?	Web access/error records plus host audit/process evidence	The linking identifiers or timestamps must distinguish execution from a failed or ordinary request.
Which identity and host account for remote access?	VPN/session records, authentication and host events	Identity and session links must be observable; do not infer authorization from success alone.
Is unusual DNS activity consistent with data transfer?	DNS records plus endpoint activity; optional PCAP	Without process or content evidence, limit the conclusion to the observable pattern.
Was a privilege change followed by execution?	Authentication, audit and process events	Require ordered evidence of the change and affected process, not an attack label.

The evaluator rubric lists the admissible answer set, three to six mandatory atomic findings, alternative valid evidence paths, material unsupported claims, required source IDs, and observability limits. Original attack labels and operator logs establish scenario facts but are never provided to agents. Two reviewers validate what the visible evidence actually warrants. A correct malicious label without an adequate visible evidential basis fails ESR. Where multiple verdicts are defensible, use a set-valued rubric and omit ordinary one-hot Brier scoring unless a unique epistemic category is adjudicated.

Assign each case a source environment, campaign ID, mechanism group, recording hash, temporal bounds, and overlap group. Cases sharing records, a causal chain, or simulation configuration form a cluster. Do not obtain nominal sample size by generating many overlapping windows. Freeze a small number of question types per episode and report both the number of questions and the number of independent source clusters.

C4 Evidence preparation and tool interface

Preserve raw records and create normalized Parquet or SQLite tables with record ID, original file hash and line/offset, timestamp and time zone, source type, entity identifiers, event type, and raw content. Normalization must preserve missing values and parser failures; never silently drop contradictory events. The evaluator keeps the private mapping to original filenames and annotation files. AIT line-based annotations must be linked before filtering or reordering records [16].

Use one read-only query service for all conditions: list_sources, describe_schema, search_events, get_records, and aggregate_events. Return deterministic ordering, explicit truncation flags, and cursor pagination. Limit each page to 200 records and 12,000 characters initially; tune only on development data. Ground-truth lookups, network access, unrestricted filesystem reads, and public search are unavailable during evaluation. A/H use these fixed operations; AS/HS may additionally run code on the same mounted evidence, with every scanned file and resource cost logged.

Script access can read the complete permitted case snapshot, not merely a selected evidence bundle; both AS and HS receive identical mounts. A/H may page through the same data via the query interface. In fixed-bundle replay all methods receive exactly the same bundle and have no additional retrieval or code access. A fixture checker may test schemas, joins, timestamp units and duplicate handling without exposing incident labels. Independently recompute sampled script findings for grading; never let the script author's own test be the sole correctness authority.

For OpTC, retain object identifiers and apply a deterministic, documented treatment of duplicate process objects and device/drive path forms. Do not merge processes solely because executable names match. Its published errata also note missing hashes for module loads and a flow acuity issue [19]. If a required edge cannot be resolved, exclude that question or mark the corresponding finding unresolved; do not repair telemetry using the secret answer.

C5 Recorded-data interventions

Version	Construction	Validity check
V0	Full permitted case snapshot	Mandatory findings have a visible support path.
V1	Bijective renaming of users, hosts, IPs and non-semantic identifiers across all sources	Preserve relations, subnet roles and operationally meaningful values; audit embedded strings.
V2	Expose additional unrelated same-environment records; target twice the original record count	No causal or evidential link to the focal question; unchanged correct findings.
V4	Hide a decisive source or event set and all redundant routes to the selected mandatory finding	Two plausible answers remain consistent with every accessible tool result.
V5	Expose common incomplete bundle, elicit answer, release fixed decisive evidence	Release resolves a pre-specified uncertainty; no fabricated log records.

A V1 case is eligible only if the mapping preserves operational meaning; do not change a command whose semantics is the evidence. V2 adds a realistic retrieval burden, so repeat a subset with token-matched context to separate distractor content from context length. Sample distractor windows without accessing agent performance. Do not splice records from different environments into a supposedly natural incident. The proposed 2× count is a target, not a requirement that overrides record quality.

For V4, model the admissible evidence paths for the chosen finding, then remove a cut set that blocks every path across all tools, copies, indexes and derived features. Re-run reference queries against the restricted snapshot. If alternative evidence still resolves the case, the variant is invalid. The required response identifies the missing link while retaining other supported facts; blanket abstention need not pass.

For V5, use a fresh two-stage run with 20 calls, 15,000 generated tokens and ten minutes per stage, totaling the ordinary cap. Supply the shared bundles at both checkpoints; count their input tokens. The incomplete bundle may support uncertainty rather than an incorrect verdict. Report justified resolutions, corrections and harmful reversals separately. Add a development-frozen 20-case late-distractor control if resources permit, outside the core run ceiling. Before release, every query and script mount exposes only the first-stage evidence; the hidden bundle and equivalent records cannot be retrieved through another path. Unlock the same additional records for every condition at the checkpoint.

C6 Experiments and run allocations

Experiment	Comparison and outcome	Target runs
E1 Core investigation	A, H, AS, HS; V0 ESR primary, V1/V2/V4/V5 secondary	80 × 5 × 4 × 2 × 3 = 9,600 maximum

Experiment	Comparison and outcome	Target runs
E2 Common evidence	A-style vs H-style synthesis; independently collected V0 bundles; ESR	80 × 2 × 2 × 3 = 960
E3 Script validation	HS vs HS without validation on 40 script-useful V0 cases; ESR	240 new runs; reuse 240 matched HS runs from E1
E4 External transfer	A, H, AS, HS on 24 OTRF + 24 OpTC V0 cases; source-specific ESR	48 × 4 × 2 × 3 = 1,152
Total planned ceiling	Excludes pilots, collectors, optional controls and regeneration	11,952 agent/synthesis runs

E1 answers whether explicit hypotheses help without scripting (H versus A), and whether the full approach helps when code access is matched (HS versus AS). The four arms use the same facts and budgets. Keep H and A matched for lesson content: H's primary structural comparison adds graph/action contracts but no extra lessons. HS includes the full method; its advantage is a package effect unless ablations isolate components. Measure actual resource use, not just maximum allowances.

E2 uses a deterministic development-frozen collector independent of the evaluated policies. The collector must search with task scope and schemas, not test labels. Its acquisition costs are reported separately. A reviewer-built oracle bundle is a distinct diagnostic, not the principal replay. If a bundle omits decisive evidence, the reference answer is recalculated for that bundle; do not reward guessing the full-record verdict. Preserve identical order and evidence content across synthesis policies.

E3 selects cases before agent runs based on a rubric requiring useful joining, aggregation or reconstruction. Disable only the validation gate; preserve prompts, script tools and aggregate budget as closely as possible. Report execution success, independently verified findings and false findings for all attempted scripts, including zero-yield cases. If fewer than 40 cases qualify, use all eligible cases and revise the count before running agents. The matched HS comparator must use the same case and run indices as E1.

E4 freezes prompts and learned lessons after AIT development. Select OTRF cases across at least six source-supported mechanism groups where available, such as credential access, remote execution and persistence; the catalogue includes LSASS, WMI, DCSync and scheduled-task recordings [18]. Group related captures together and report the actual number of independent recordings. For OpTC, curate provenance questions from documented episodes and report campaign clustering. Do not claim 24 independent campaigns or infer benign outcomes from unlabeled telemetry. Source-level scores are not pooled into the AIT primary estimate.

Secondary R/P controls use a preregistered 20-case V0 subset stratified across the four test environments: two additional arms × 20 cases × two backbones × three repeats = 240 new runs. Compare with A/H outcomes on exactly that subset. A no-evidence model diagnostic uses the same questions and common facts but no incident records; score unsupported specificity as well as answers. Its 480 runs (80 × 2 × 3) are additional and measure possible prior/template dependence, not a proof of contamination.

C7 Optional genuine counterfactual study

Existing recorded logs cannot answer what would have happened under a changed incident mechanism without additional modeling assumptions. Therefore V3 is optional and separately reported. Use the Kyoushi environment [17] as a starting point for newly recorded pairs, not as a claim that such pairs already exist. Pin infrastructure and simulation versions, restore the same initial snapshot, change one specified mechanism, and collect both members in full. Background workload seeds are shared where feasible, with residual nondeterminism measured.

Target 24 pairs from at least six mechanism templates, subject to feasibility. Illustrative pair: a web request that leads to execution versus a matched attempted request that fails because the vulnerable path is disabled. The expected conclusion changes from supported execution to an unsuccessful attempt, not automatically to benign activity.

Validate all dependent host and network records; do not change only a verdict or a line of telemetry. Missing downstream traces must be consistent with the modified mechanism.

Run A and H on both members, with two backbones and three repeats: 24 × 2 × 2 × 2 × 3 = 576 additional runs. Use CPS, evidence-supported mechanism discrimination and per-template effects. Templates and repeated workloads remain clusters; six templates still constrain generalization. This is a newly generated extension anchored in open-source infrastructure, explicitly distinct from experiments using existing datasets only.

C8 Scoring, statistics and completion criteria

Freeze ESR rubrics before runs. An ESR pass requires an admissible conclusion, all mandatory findings, valid source support and no material unsupported claim. Blinded reviewers score shuffled standardized outputs; they do not see condition names or hypothesis-contract formatting. Use automated checks for identifiers, timestamps and cited record existence, but human review for entailment. Report agreement and adjudication counts. More verbose answers receive no extra credit.

The four confirmatory contrasts are H−A V0 ESR, structured−standard fixed-bundle ESR, HS−AS V0 ESR, and HS−unvalidated HS V0 ESR on the eligible script subset. Report paired effect sizes in percentage points and both case-conditional and environment-level uncertainty. All estimates preserve repeat pairing and source clusters. Power simulation uses development estimates of discordance and cluster correlation; publish the detectable effect rather than assuming the nominal run count yields adequate power. Four test environments support a finite-corpus comparison, not a confident population claim.

Report separate panels for dataset, backbone and variant, with N cases, N source clusters and N runs beside every estimate. Show malicious, benign and insufficient-evidence strata. Report confidence calibration only for uniquely adjudicated labels; use coverage and selective risk alongside abstention. A compact figure set comprises V0 ESR against cost, variant pair success and missing-evidence ESR, provisional-to-final revision outcomes, and matched component/replay effects. Keep all result tables blank until measured.

The first pilot uses ten development cases with V0 and eligible V4, A and H, one backbone and two repeats: at most 80 runs. A second ten-case script pilot with AS and HS adds 40 V0 runs. Pilot costs determine affordability and help freeze caps; do not replace planned case clusters with extra repeats. Archive data preparation cost separately from per-investigation inference cost. No external dataset downloads or model experiments were performed as part of writing this protocol.

Before execution, publish or register: source versions and checksums; archive and license inventory; train/development/test manifest; case and cluster IDs; raw-to-normalized mappings; transformation code and seeds; permitted tools; private evaluator rubric hashes; frozen prompts and lessons; model versions; resource limits; and analysis scripts. After execution, release permissible raw subsets or reproducible download/extraction recipes, anonymized traces, adjudicated findings, deviations and complete denominators. Preserve source license terms, including AIT's noncommercial share-alike conditions; release original benchmark code under a separately stated compatible license.